Guava Fruit Disease Classification

Custom CNN from scratch vs. MobileNetV2 transfer learning (GFDD24)

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Course project — seed 42 everywhere; CPU-only build (Apple Silicon, 224px)

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Agenda

- Problem and motivation
- Related work: overview table (2.1), method families (2.2), gap (2.3)
- Dataset audit + leakage finding
- CNN and transfer-learning concepts; data pipeline
- Model A vs Model B; project structure and tools
- Results, confusion matrices, Grad-CAM
- Error analysis, limitations, conclusion, references

Problem and motivation

- Classify guava fruit photos into **Anthracnose / Fruit Flies / Healthy** to enable early, automatic disease triage.
- **Model A**: custom 4-block CNN from scratch (1.24M params) vs. **Model B**: ImageNet-pretrained MobileNetV2 + head, then fine-tune last 30 layers.
- Released splits leak (same fruit in train and test) — so we report **released and leak-free** accuracy side by side, plus Grad-CAM evidence.
- Headline (test, seed 42): A **0.911** / F1 0.917 vs. B **0.992** / F1 0.991.

Related work (2.1): comparative overview

Study	Method	Data	Result	Limit
Mostafa et al. 2021	5 CNNs, R-50 best	321→2889, 5 cl.	99.54%	other set, no ctl/XAI
Doutoum et al. 2023	Transfer, E-B3	1834 leaves, 5 cl.	94.93%	leaf-only, no XAI
Nandi et al. 2022	5 CNNs + quant.	681+8525, 5 cl.	99% / 97%	mixed, no XAI
Kilci & Koklu 2024	SqueezeNet+GB	GFDD24 3784	95.6%	curated only
Tewari et al. 2024	DenseNet169	527, 5 cl.	99.62%	small, no XAI

- Full 12-study table with verification URLs: `report/survey.md` §8; BibTeX: `report/references.bib`. Pratama 2025 dropped (unverifiable).

Related work (2.1 cont. / 2.2): families

Study	Method	Data	Result	Limit
Zhou 2026	VGG16+MNetV2+Grad-CAM	527, 5 cl.	96%	other taxonomy
ResNet-101 2025	Transfer+XAI	GFDD24	98.48%	curated only
CNN-GB ens. 2025	EffNet-B0+XGB	GFDD24	∼99.99%*	preprint, leaked
Mohanty 2016	Alex/GoogLeNet	PlantVillage 54k	99.35%	lab-field gap

Method families (2.2)

Transfer architectures (ours B) / hybrid deep-features+boosting / data-centric (aug-before-split, quant., CLAHE).

Related work (2.3): gap and positioning

Defensible gap

No verified GFDD24 study reports **(i)** from-scratch vs two-phase transfer head-to-head, **(ii)** with duplicate quantification, **(iii)** a leak-free control, **(iv)** Grad-CAM on the same split. [5,8,9] report GFDD24 accuracy (95.6–99.99%) on the leaked split without controls.

- We do not claim "no MobileNetV2 on guava" ([4,7] did). Claim softens to "among the first" if a 4-part study surfaces — table gives everything to check.

Dataset (GFDD24, Mendeley DOI 10.17632/bkdkc4n835.1)

- 473 field originals (Rajshahi/Pabna, July, pathologist-verified), unsharp+CLAHE to 512x512 RGB PNG, augmented x8 to **3,784 images**.
- On disk: train 2647 (1080 / 918 / 649), val **755** (308 / 262 / 185), test 382 (156 / 132 / 94).
- Release notes say val 775 — **20-image shortfall reported, not hidden**.
- All 3,784 decode clean (0 corrupt); folders read from disk: Anthracnose, fruit_fly, healthy_guava.

Concepts: CNN classifier

- Conv 3×3 learns local lesion/puncture textures; stacked blocks (32/64/128/256) build scale; MaxPool halves resolution; GAP averages to 256-d.
- BatchNorm stabilizes layer inputs (Ioffe & Szegedy 2015); Dropout 0.25/0.5 breaks co-adaptation (Srivastava et al. 2014).
- Softmax + categorical cross-entropy; Adam 1e-3 (Kingma & Ba 2015). Fix found here: BN momentum 0.9 (not 0.99) so moving averages track fast-learning small-data stats.

Concepts: two-phase transfer learning

- Yosinski et al. 2014: early filters (edges/blobs) transfer; late layers specialize. ImageNet (Deng et al. 2009) is the source.
- Phase 1 (head warm-up): backbone frozen incl. BatchNorm, lr 1e-3, 15 epochs. Phase 2: unfreeze last 30 layers, BN kept frozen, lr 1e-5, 20 epochs.
- MobileNetV2 (Sandler et al. 2018): inverted residuals + linear bottlenecks; inputs mapped $[0, 1] \to [-1, 1]$ via Rescaling (no Lambda, safe .keras reload).

Data pipeline

- `image_dataset_from_directory`, categorical labels, class order from disk, seed 42, shuffle train only; rescale $[0, 1]$; prefetch.
- Train-only augmentation: flip H+V, rotation 0.15, zoom 0.15, contrast 0.1, brightness 0.1 (value_range fixed to $(0, 1)$ — default $(0, 255)$ blew train to 24).
- 224px CPU fallback (native 512px); batch 32; single seed 42 (stated scope cut).

Model A vs Model B

A: custom CNN (1,243,683 params)

- 4×[Conv-ReLU-BN x2, Pool, Drop 0.25]; GAP; Dense256-ReLU-BN-Drop0.5; softmax. Adam 1e-3, $\leq$40 ep.

B: MobileNetV2 TL (2,586,691)

- Frozen backbone + Dense256-Drop0.4 head; FT last 30 (BN frozen). 1e-3/15 + 1e-5/20.

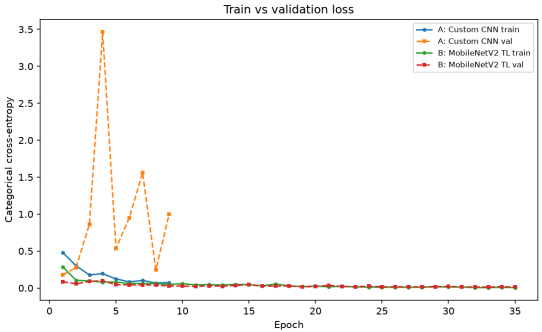
- Shared: CCE, ES (val_loss, patience 8, restore best), ReduceLROnPlateau, best-val_acc checkpoint, save/reload parity True.

Project structure and tools

- `config.py data_pipeline.py models.py train.py evaluate.py scripts/{audit,gradcam,make_figures}.py`
- `Dataset/ Dataset_leakfree/ outputs/{models,logs,plots,audit,results}/ report/`
- Tools: TF 2.21 CPU, sklearn, matplotlib/seaborn, imagehash, Tectonic Beamer. One command per stage in README.

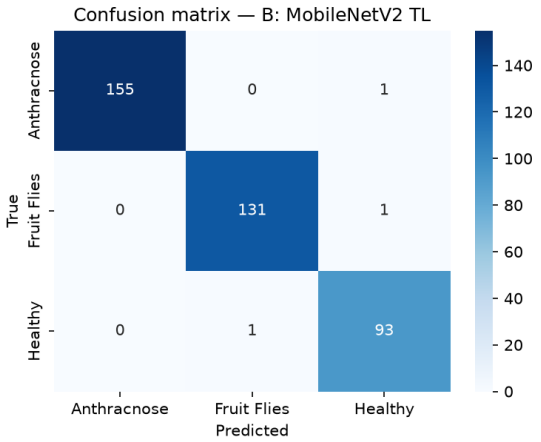
Results (test 382, seed 42) — every number from executed runs

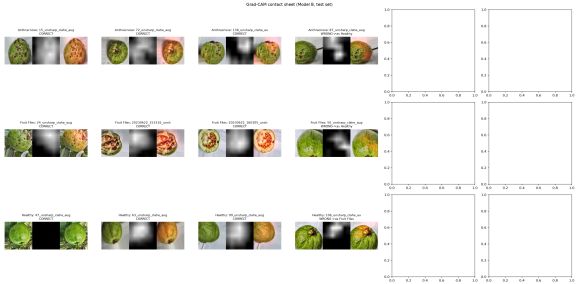
Model	Test	F1	Val
A: Custom CNN	0.9110	0.9173	0.9364
B: MobileNetV2	0.9921	0.9911	0.9960
B frozen	0.9948	0.9942	0.9907
B no-augment	0.9948	0.9946	0.9974



- B beats A by ∼8pp; frozen/noaug match full B on leaked split (head suffices; release aug covers variation). Leak-free: A-control 0.9923 (partial ckpt), B-control 0.9872 — no visible inflation (task saturates).

Confusion matrices and Grad-CAM





- B: 3 errors total. A: 26 Anthracnose→Fruit-Flies. Grad-CAM (Selvaraju et al. IJCV 2020) on deepest backbone conv: lesions/punctures highlighted, not background.

Error analysis and limitations

- Confusable pair: Anthracnose lesions vs fruit-fly punctures (dark spots). Curves: A wobbles (val 0.93→0.71→0.94, ES kept ep8); B tracks cleanly.
- Leakage: 454/473 originals span splits; 100% val/test share an original with train — released scores are near-duplicate recognition (Tampu precedent). Control reported alongside.
- Limits: single seed 42; 224px fallback; val 755 vs 775; leakfree-custom ckpt interrupted 17/40 (best kept, stated).

Conclusion and future work

- Transfer (B, 0.992) beats from-scratch (A, 0.911) on 2.6k images, faster (14.5 vs 31.6 min) and stabler. Frozen head alone already 0.995 on leaked split.
- Honest reporting (leakage quantified + control + Grad-CAM) matters more than the last point of accuracy.
- Future: field photos from new orchards; 512px + multi-seed; EfficientNetB0/ResNet50 backbones (already supported); deployment-size quantization (Nandi precedent).

References

1 Al Amin et al. GFDD24, Mendeley Data 2024, DOI 10.17632/bkdkc4n835.1.

2 Mostafa et al. Appl. Sci. 12(1):239, 2021. [3] Doutoum et al. WSEAS 20:356-363, 2023. [4] Nandi et al. MIET/arXiv:2209.12557, 2022.

5 Kilci & Koklu, Proc. FIAR 2024. [6] Mohanty et al. Front. Plant Sci. 2016. [7] Sandler et al. CVPR 2018.

8 Tampu et al. Sci. Data 9:580, 2022. [9] Selvaraju et al. IJCV 2020. [10] Kingma & Ba ICLR 2015; Ioffe & Szegedy ICML 2015; Yosinski et al. NeurIPS 2014. Full 22-entry BibTeX: `report/references.bib`.

Thank you

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Code, results, and slides fully reproducible from `README.md` (seed 42).